

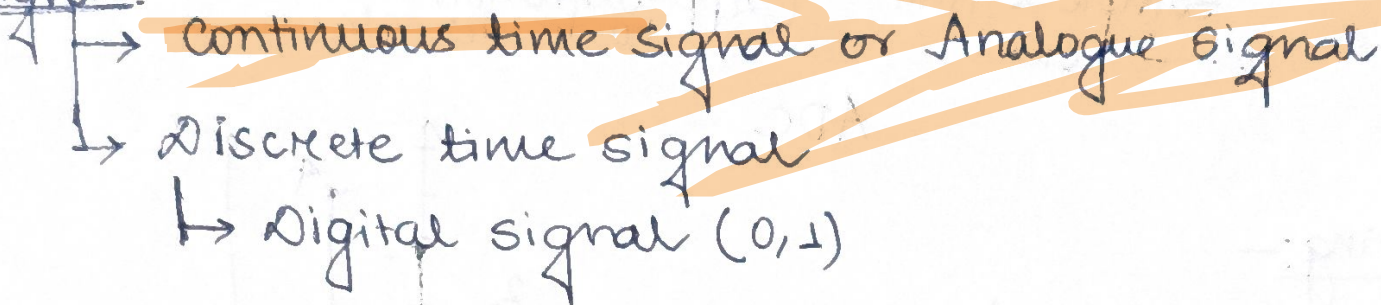
Digital Signal Processing (DSP)

- A signal is a physical quantity that varies with two or more independent variables and carries information.

What is signal processing?

⇒ The process of operation in which the characteristics of a signal amplitude, shape, phase, frequency etc undergoes a change to extract useful information carried by the signal is called signal processing.

Signals:



* 1) Analog signal processing (ASP):-

modification of analog signals to extract useful information

e.g: ASP → filters, frequency multipliers.

2) Digital signal processing (DSP):-

Digital representation of signals and the use of digital processors to analyze, modify or extract information from signals.

⇒ Advantages of DSP over ASP:-

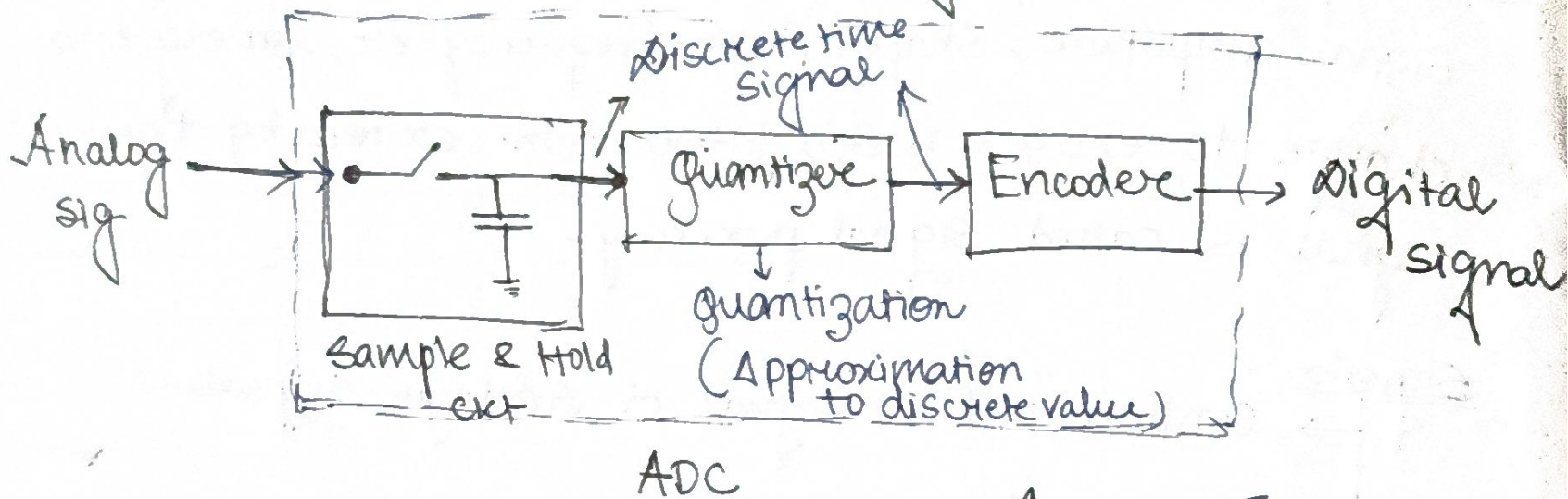
- Accuracy
- Flexibility
- Superior performance
- Adaptability
- Reproducibility
- Data logging

⇒ Disadvantages:-

- Increased system complexity
- Speed & cost
- Finite wordlength

→ For transmission of signal over a long distance high frequency carrier signal is required by transmitter.

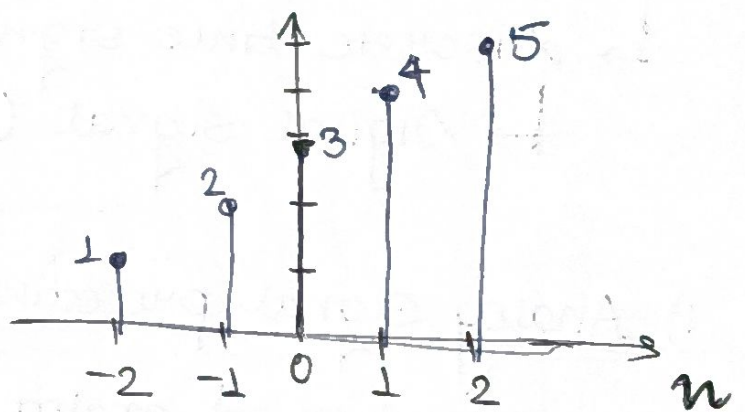
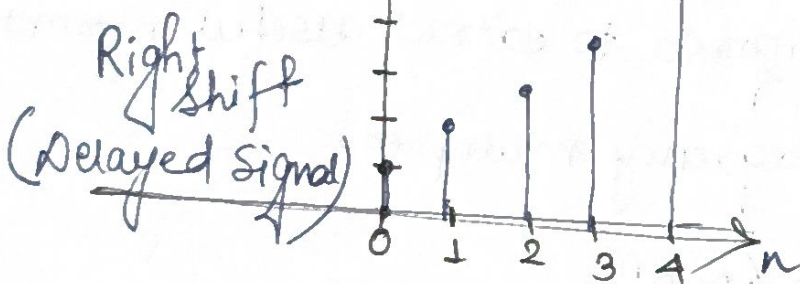
The receiver receives the high frequency signal and extracts the useful information using low-pass filter.



* Shifting:-

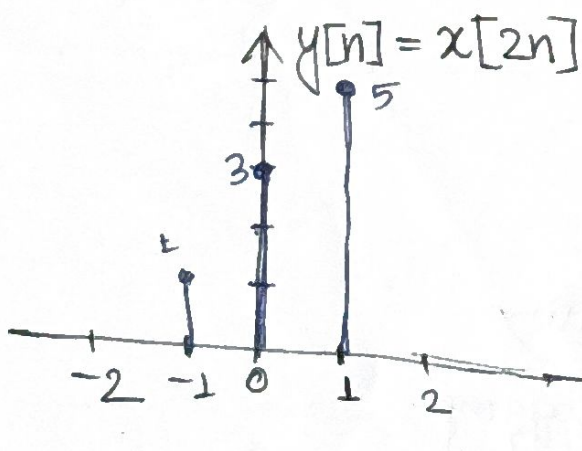
$$x[n] = \{1, 2, 3, 4, 5\}$$

$$y[n] = x[n-2]$$

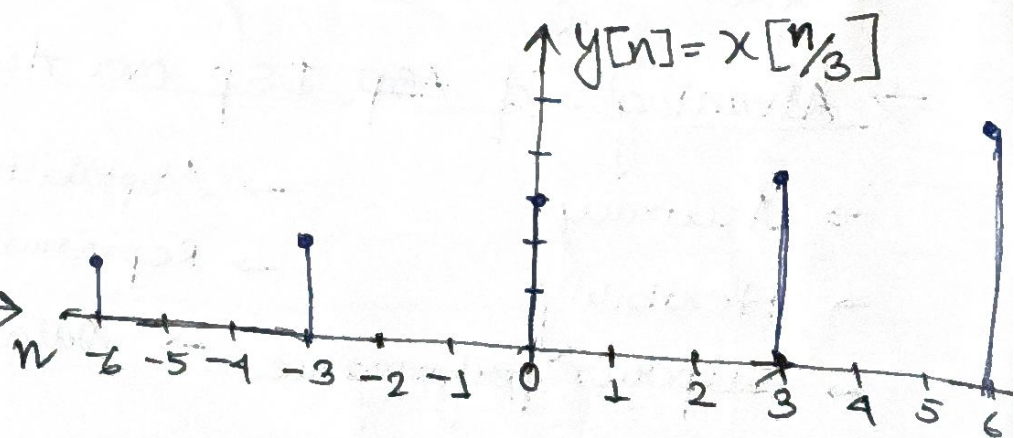


* Scaling:-

(a) Decimation:

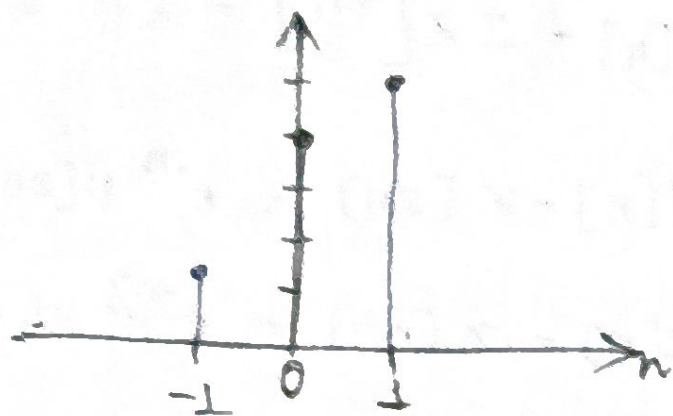


(b) Interpolation

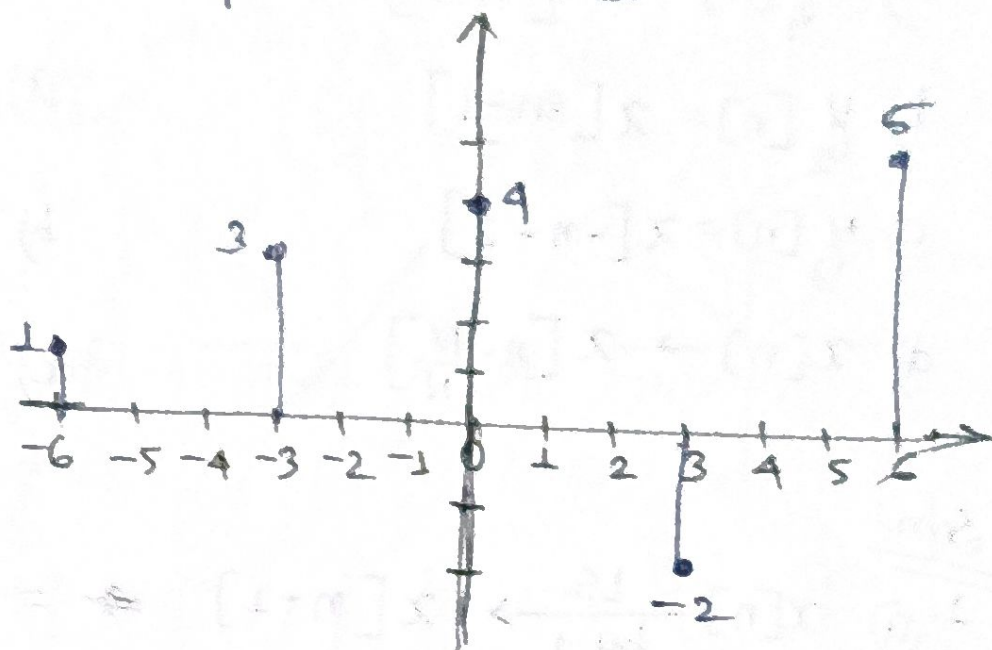


Q. $x[n] = \{1, 3, 4, -2, 5\}$

a) $y[n] = x[2n]$



b) $z[n] = x[n/3]$



c) $x[n] \rightarrow x\left[-\frac{n}{m} - k\right] =$

① $x[n] \xrightarrow{\text{Ref}} x[-n]$

② $x[-n] \xrightarrow[\text{by } k]{\text{LS.}} x[-(n+k)] = x[-n-k]$

③ $x[-n-k] \xrightarrow[\text{by } m]{\text{Int.}} x\left[-\frac{n}{m} - k\right]$

d) $x[n] \rightarrow x[-n-k] =$

① $x[n] \xrightarrow[\text{by } k]{\text{RS}} x[n-k]$

② $x[n-k] \xrightarrow{\text{Ref}} x[-n-k]$

e) $x[n] \rightarrow x[-n+k]$

① $x[n] \xrightarrow[\text{by } k]{\text{LS}} x[n+k]$

or ① $x[n] \xrightarrow{\text{Ref}} x[-n]$

② $x[n+k] \xrightarrow{\text{Ref}} x[-n+k]$

② $x[n] \xrightarrow[\text{by } k]{\text{LS}} x[-(n-k)] = x[-n+k]$

$$x[n] = \{1, 2, 3, 4, 5, 6\}$$

↑

1.

a) $y_1[n] = x[-n+1]$

b) $y_2[n] = x[3n-1]$

c) $y_3[n] = x[-n-2]$

d) $x[n] \rightarrow x[n - \frac{M}{N}]$

2. $x[n] = e^{-n/2} u[n]$

a) $H[n] = 2x[\frac{5n}{3}] = 2e^{-\frac{5n}{6}} u[\frac{5n}{3}]$

b) $y[n] = x[2n] = e^{-n} u[2n]$

c) $z[n] = x[n^2] = e^{-n^2/2} u[n^2]$

Soln:

1. a) $x[n] \xrightarrow[\text{by 1}]{\text{L.S.}} x[n+1] = \{1, 2, 3, 4, 5, 6\}$

$x[n+1] \xrightarrow{\text{Ref.}} x[-n+1] = \{6, 5, 4, 3, 2, 1\}$

b) $x[n] \xrightarrow[\text{by 1}]{\text{R.S.}} x[n-1] = \{1, 2, 3, 4, 5, 6\}$

$x[n-1] \xrightarrow[\text{by 3}]{\text{Decm}} x[3n-1] = \{0, 2, 5, 0\}$

c) $x[n] \xrightarrow[\text{by 2}]{\text{R.S.}} x[n-2] = \{1, 2, 3, 4, 5, 6\}$

$x[n-2] \xrightarrow{\text{Ref.}} x[-n-2] = \{6, 5, 4, 3, 2, 1\}$

d) $x[n] \xrightarrow[\text{by } \frac{M}{N}]{\text{R.S.}} x[\frac{n-M}{N}] \xrightarrow[\text{by } MN]{\text{R.S.}} x[\frac{n-MN}{N}]$

2. $x[n] = e^{-n/2} u[n]$

d) $x[n] \xrightarrow[\text{by } N]{\text{Inter.}} x[\frac{n}{N}] \xrightarrow[\text{by } M]{\text{R.S.}} x[\frac{n-M}{N}] \xrightarrow[\text{by } N]{\text{Dec.}} x[n - \frac{M}{N}]$

1. a) Deterministic:-

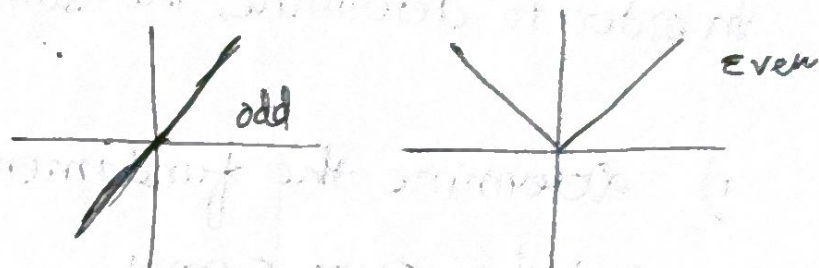
$$x(t) = t, x[n] = n$$

$$y(t) = \sin \omega t, y[n] = \sin \omega n$$

2. Even & odd signal:-

$$x[-n] = x[n] \Rightarrow \text{Even Signal}$$

$$x[-n] = -x[n] \Rightarrow \text{odd signal}$$



$$E[x(n)] = \frac{x[n] + x[-n]}{2}$$

$$O[x(n)] = \frac{x[n] - x[-n]}{2}$$

H/w

① Find out the Even & odd sequence of unit step & ramp sign

Eg: $x[n] = \{1, 2, 1, 5, -2\}$
 $\uparrow$
 $n=0$

	$n = -2$	$n = -1$	$n = 0$	$n = 1$	$n = 2$
$x[n]$	1	2	1	5	-2
$x[-n]$	-2	5	1	2	1
$E[x(n)]$	$-\frac{1}{2}$	$\frac{7}{2}$	1	$\frac{7}{2}$	$-\frac{1}{2}$
$O[x(n)]$	$\frac{3}{2}$	$-\frac{3}{2}$	0	$\frac{3}{2}$	$-\frac{3}{2}$

H/w:-

$x[n]$

* Periodicity

$$e^{j\omega n} = \cos \omega n + j \sin \omega n = e^{j\omega(n+N)} = e^{j\omega n} e^{j\omega N}$$

$$e^{j\omega N} = \cos \omega N + j \sin \omega N = \cos 2\pi K + j \sin 2\pi K \quad \omega = 2\pi \frac{1}{T}$$

$$\therefore \omega N = 2\pi K$$

$$\Rightarrow \frac{\omega N}{2\pi} = \frac{K}{N}$$

$$\Rightarrow T = \frac{2\pi}{\omega} = \frac{N}{K}$$

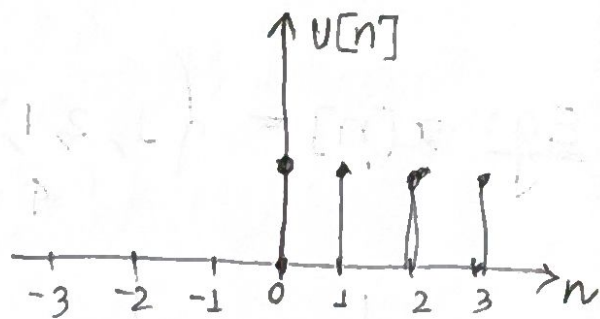
$$\Rightarrow T = \frac{2\pi}{\omega}$$

In order to determine, the sum of two periodic signal is periodic or not

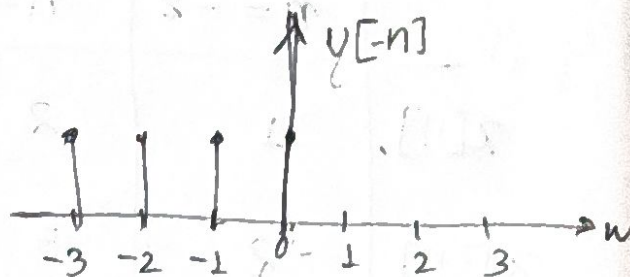
- 1) determine the fundamental period of the individual signal and the sum signal.
- 2) Find the ratio of fundamental period of ~~the~~ 1st signal with other signal.
- 3) If the ratio's are rational, then the sum is also rational.

* H/W:

$$1. x[n] = u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

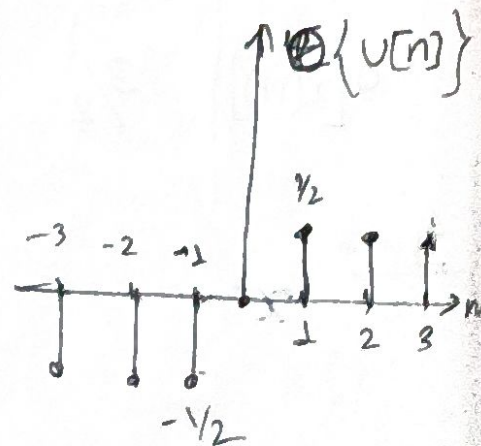


$$x[-n] = u[-n] = \begin{cases} 1, & n \leq 0 \\ 0, & n > 0 \end{cases}$$



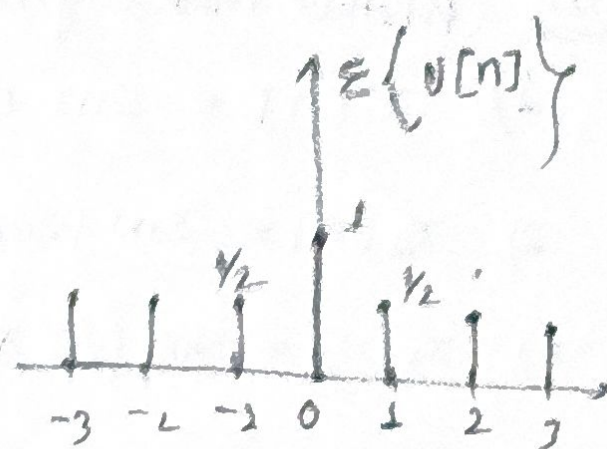
$$\therefore \Theta\{u[n]\} = \frac{x[n] + x[-n]}{2}$$

$$= \begin{cases} -\frac{1}{2} u[n], & n \leq -1 \\ 0, & n = 0 \\ \frac{1}{2} u[n], & n \geq 1 \end{cases}$$

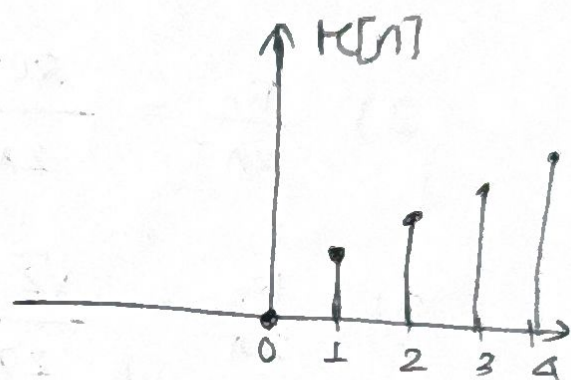


$$\therefore E\{v[n]\} = \frac{v[n] + v[-n]}{2}$$

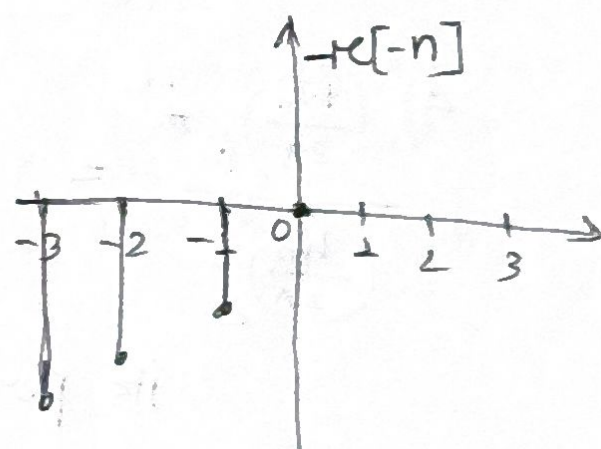
$$= \begin{cases} \frac{1}{2} v[n], & n \leq -1 \\ 1, & n = 0 \\ \frac{1}{2} v[n], & n \geq 1 \end{cases}$$



$$2. \quad x[n] = \kappa[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

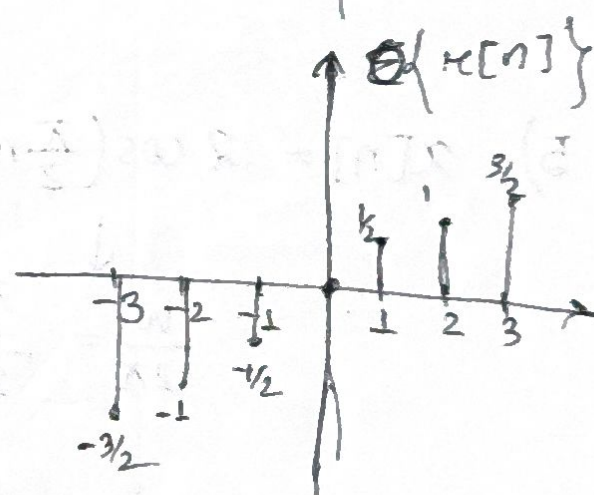


$$\therefore x[-n] = -\kappa[-n] = \begin{cases} -n, & n \leq 0 \\ 0, & n > 0 \end{cases}$$



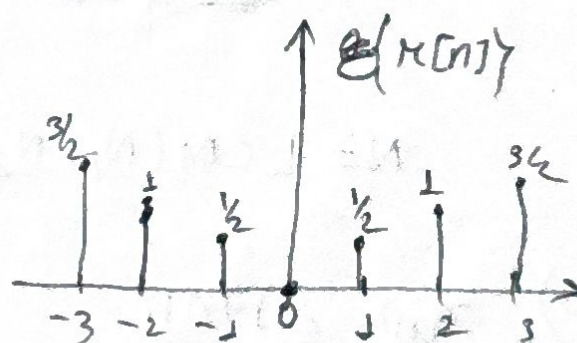
$$\therefore E\{\kappa[n]\} = \frac{\kappa[n] + \kappa[-n]}{2}$$

$$= \begin{cases} \frac{1}{2} n, & n \geq 0 \\ 0, & n = 0 \\ -\frac{1}{2} n, & n < 0 \end{cases}$$



$$\therefore E\{\kappa[n]\} = \frac{\kappa[n] - \kappa[-n]}{2}$$

$$= \begin{cases} \frac{1}{2} n, & n \geq 0 \\ \frac{1}{2} n, & n = 0 \\ -\frac{1}{2} n, & n < 0 \end{cases}$$



Q. whether these signals are periodic.

1) $x_1[n] = \sin 15\pi n$

4) $x_4[n] = x_1[n] + x_2[n]$.

2) $x_2[n] = \sin(20\pi n)$

its periodicity.

3) $x_3[n] = \sin(\sqrt{2}\pi n)$

Soln.

1) $\frac{\omega_n}{2\pi} = \frac{15\pi}{2\pi} = \frac{15}{2} \rightarrow \text{periodic} \quad \frac{K}{N}$

2) $\frac{\omega_n}{2\pi} = \frac{20\pi}{2\pi} = \frac{10}{1} \rightarrow \text{periodic}$

3) $\frac{\omega_n}{2\pi} = \frac{\sqrt{2}\pi}{2\pi} = \frac{1}{\sqrt{2}} \rightarrow \text{Non periodic}$

4) $\frac{\left(\frac{\omega_n}{2\pi}\right)_1}{\left(\frac{\omega_n}{2\pi}\right)_2} = \frac{15/2}{10/1} = \frac{15}{20} = \frac{3}{4} \rightarrow \text{periodic.}$

its period, $N = \text{LCM}(2, 1) = 2$

5) $x[n] = 2 \cos\left(\frac{\pi}{2}n\right) \sin\left(\frac{\pi}{8}n\right) \Rightarrow \sin\left(\frac{\frac{\pi}{2} + \frac{\pi}{8}}{2}n\right) + \sin\left(\frac{\frac{\pi}{2} - \frac{\pi}{8}}{2}n\right)$

$\therefore \frac{\omega_1}{2\pi} = \frac{\pi/2}{2\pi} = \frac{1}{4} \quad \frac{\omega_2}{2\pi} = \frac{\pi/8}{2\pi} = \frac{1}{16} \rightarrow \text{periodic}$

$\therefore \text{LCM} = N_1 = 4$

$N_2 = 16$

$N = \text{LCM}(N_1, N_2) = 16$

6) $x[n] = e^{j7\pi n}$

$\frac{\omega_6}{2\pi} = \frac{7\pi}{2\pi} = \frac{7}{2}, N = 2 \text{ samples}$

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 \rightarrow \text{Energy signal, } \underbrace{0 < E_x < \infty}_{\text{finite}}, P_x = 0$$

$$P_x = \frac{1}{(2N+1)} \sum_{n=-N}^N |x(n)|^2 \rightarrow \text{power signal, } \underbrace{0 < P_x < \infty}_{\text{finite}}, E_x = \infty$$

Q. Nature of signal:

① $x_1(n) = \left(\frac{1}{2}\right)^n u(n)$

③ $x_3(n) = \cos \frac{\pi}{4} n$

② $x_2(n) = e^{j\left(\frac{\pi}{2}n + \frac{\pi}{8}\right)}$

Soln:

① $E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-\infty}^{\infty} \left| \left(\frac{1}{2}\right)^n u(n) \right|^2 \quad \left(\frac{a}{1-r}\right)$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2n} = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$$

$$= \boxed{\text{Diagram of a geometric series sum}} \cdot \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

$$P_x = \frac{1}{(2N+1)} \sum_{n=-N}^N \left| \left(\frac{1}{2}\right)^n u(n) \right|^2$$

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N \left(\frac{1}{4}\right)^n u(n)$$

$\Rightarrow$ It is a Energy signal

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=0}^N \left(\frac{1}{4}\right)^n$$

$$= 0$$

$$b) x_2(n) = e^{j(\frac{\pi}{2}n + \pi/8)}$$

$$E_x = \sum_{n=-\infty}^{\infty} |e^{j(\frac{\pi}{2}n + \pi/8)}|^2$$

$$= \sum_{n=-\infty}^{\infty} 1^2$$

$$= \infty$$

$$P_2 = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |e^{j(\frac{\pi}{2}n + \pi/8)}|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N 1$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} (2N+1) = 1 \Rightarrow \text{Power signal.}$$

$$c) x_3(n) = \cos \frac{\pi}{4} n$$

$$E_x = \sum_{n=-\infty}^{\infty} |\cos \frac{\pi}{4} n|^2 = \sum_{n=-\infty}^{\infty} \cos^2 \left(\frac{\pi}{4} n \right) = \sum_{n=-\infty}^{\infty} \left[\frac{1 + \cos \left(\frac{\pi}{2} n \right)}{2} \right]$$

$$= \sum_{n=-\infty}^{\infty} \frac{1 + \cos \left(\frac{\pi}{2} n \right)}{2}$$

$$= \infty$$

$$P_2 = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N \cos^2 \frac{\pi}{4} n$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N \left(\frac{1 + \cos \frac{\pi}{2} n}{2} \right)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \left[\frac{1}{2} (2N+1) + \sum_{n=-N}^N \frac{\cos \frac{\pi}{2} n}{2} \right]$$

$$= \frac{1}{2}$$

Power
Signal

* Linear and Non linear System:

→ A system follows both the laws of superposition (additivity) & Homogeneity (scaling) is called a linear system.

$$\text{Let, } y(n) = n x(n) \rightarrow \textcircled{i}$$

$$x_1(n) \rightarrow y_1(n) = n x_1(n) \rightarrow \textcircled{ii}$$

$$x_2(n) \rightarrow y_2(n) = n x_2(n) \rightarrow \textcircled{iii}$$

$$x_3(n) \rightarrow y_3(n) = n x_3(n) \rightarrow \textcircled{iv}$$

Assume

$$x_3(n) = a x_1(n) + b x_2(n)$$

Now, the system will be linear, if

$$y_3(n) = a y_1(n) + b y_2(n) \rightarrow \textcircled{v}$$

LHS

RHS

$$y_3(n) = n x_3(n) = n [a x_1(n) + b x_2(n)]$$

$$= a n x_1(n) + b n x_2(n)$$

$$\text{LHS} = a y_1(n) + b y_2(n) = \text{RHS}$$

∴ The system is linear.

$$\textcircled{2.} \quad y(n) = 2x(n) - 3$$

$$x_1(n) \rightarrow y_1(n) = 2x_1(n) - 3$$

$$x_2(n) \rightarrow y_2(n) = 2x_2(n) - 3$$

$$x_3(n) \rightarrow y_3(n) = 2x_3(n) - 3$$

$$\text{Assume } x_3(n) = a x_1(n) + b x_2(n)$$

$$\therefore y_3(n) = 2[a x_1(n) + b x_2(n)] - 3$$

$$= a 2x_1(n) + b 2x_2(n) - 3$$

$$y_3(n) = a y_1(n) + b y_2(n) + 3$$

Non-linear sys. //

* Time Variant & Time invariant sys.

If the output of the sys. varies/changes with time of input, called as time variant sys.

$$y(n) = n x(n)$$

$$x_1(n) \rightarrow y_1(n) = n x_1(n)$$

$$x_2(n) \rightarrow y_2(n) = n x_2(n)$$

Assume,

$$x_2(n) = x_1(n - n_0)$$

This sys. will be time invariant if

$$y_2(n) = y_1(n - n_0)$$

$$\text{LHS} = y_2(n) = n x_2(n) = n x_1(n - n_0)$$

$$\text{RHS} = y_1(n - n_0) = (n - n_0) x_1(n - n_0)$$

$\therefore \text{LHS} \neq \text{RHS} \Rightarrow \text{Time variant.}$

9. $y(n) = \sin[x(n)]$

$$x_1(n) \rightarrow y_1(n) = \sin[x_1(n)]$$

$$x_2(n) \rightarrow y_2(n) = \sin[x_2(n)]$$

Assume

$x_2(n) = x_1(n - n_0) \rightarrow \text{sys. will be time invariant if}$

$$y_2(n) = y_1(n - n_0)$$

$$\text{LHS} = y_2(n) = \sin[x_2(n)] = \sin[x_1(n - n_0)] \therefore \text{LHS} = \text{RHS}$$

$$\text{RHS} = y_1(n - n_0) = \sin[x_1(n - n_0)] \Rightarrow \text{Time invariant}$$

* Causal & Non-causal sys:-

A causal sys. is that where the o/p ~~does not~~ depend on the present & past i/p and not on future i/p.

$$y(n) = n x(n)$$

$$n = n_0$$

$$y(n_0) = n_0 x(n_0)$$

$$y(-n_0) = -n_0 x(-n_0)$$

⇒ ~~Time~~ causal sys.

$$y(n) = x(-n)$$

$$y(n_0) = x(-n_0)$$

$$y(-n_0) = x(n_0)$$

→ Non-causal sys.

* Stable & Un-stable sys:-

Stable sys. obeys BIBO rule, i.e. for Bounded input the output should be bounded.

$$\left. \begin{array}{l} |x(n)| < |B_x| < \infty \\ |y(n)| < |B_y| < \infty \end{array} \right\} \text{finite}$$

Q. $e^{x(n)}$

Soln:-

$$e^{B_x}$$

$$|x(n)| \leq B_x$$

$$-B_x \leq x(n) \leq B_x$$

$$e^{-B_x} \leq e^{x(n)} \leq e^{B_x}$$

$$e^{-B_x} \leq y(n) \leq e^{B_x}$$

Bounded o/p

H/w

invertibility & inv.
passive & lossless.
sys

$$x(0) = ?$$

$$y(n) = x(n) + x(n-1)$$

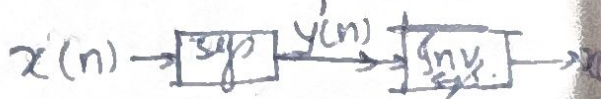
$$y(1) = x(1) + x(0)$$

} Memory sys.

* Invertibility & Inverse system

→ A system is invertible if by observing the output, we can determine its input.

For a system to ~~have~~ be invertible distinct inputs must lead to distinct outputs. So that there is one to one mapping between an input and the corresponding output.



* passive & lossless system:-

A discrete time system is said to be passive if, for every finite energy input signal $x(n)$ the output signal $y(n)$ has at most the same energy,

$$\text{i.e. } \sum_{n=-\infty}^{\infty} |y(n)|^2 \leq \sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$$

If this inequality is satisfied with an equal sign for every input signal, the discrete system is said to be lossless.

$$\text{eg:- } \sum_{n=-\infty}^{\infty} |x(n)|^2 \leq \sum_{n=-\infty}^{\infty} |y(n)|^2$$

$$\sum_{n=-\infty}^{\infty} |y(n)|^2 = \alpha^2 \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad \alpha = 1 \text{ for lossless, } \alpha < 1 \text{ for passive}$$

Q. Check stability.

$$1. \quad y(n) = r^n x(n) \quad r > 1$$

Soln: for bounded i/p. $|x(n)| \leq B_x < \infty$

$$|y(n)| = |r^n x(n)|$$

$$\leq |r^n| B_x$$

$$\leq r^n B_x$$

$$\text{for } n \rightarrow \infty \quad |y(n)| \rightarrow \infty$$

⇒ Unbounded ⇒ unstable

$$2. y(n) = \sum_{k=0}^{\infty} f^k x(n-k)$$

$$\begin{matrix} n-k=m \\ \Rightarrow n=k+m \end{matrix}$$

$$|y(n)| = \sum_{k=0}^{\infty} |f^k x(n-k)|$$

$$\leq \sum_{k=0}^{\infty} f^k B_2$$

$f > 1 \rightarrow y(n) \rightarrow \infty \rightarrow \text{unbounded} \rightarrow \text{unstable}$

$f < 1 \rightarrow y(n) \leq B_2 \rightarrow \text{stable}$

* Sampling Theorem:

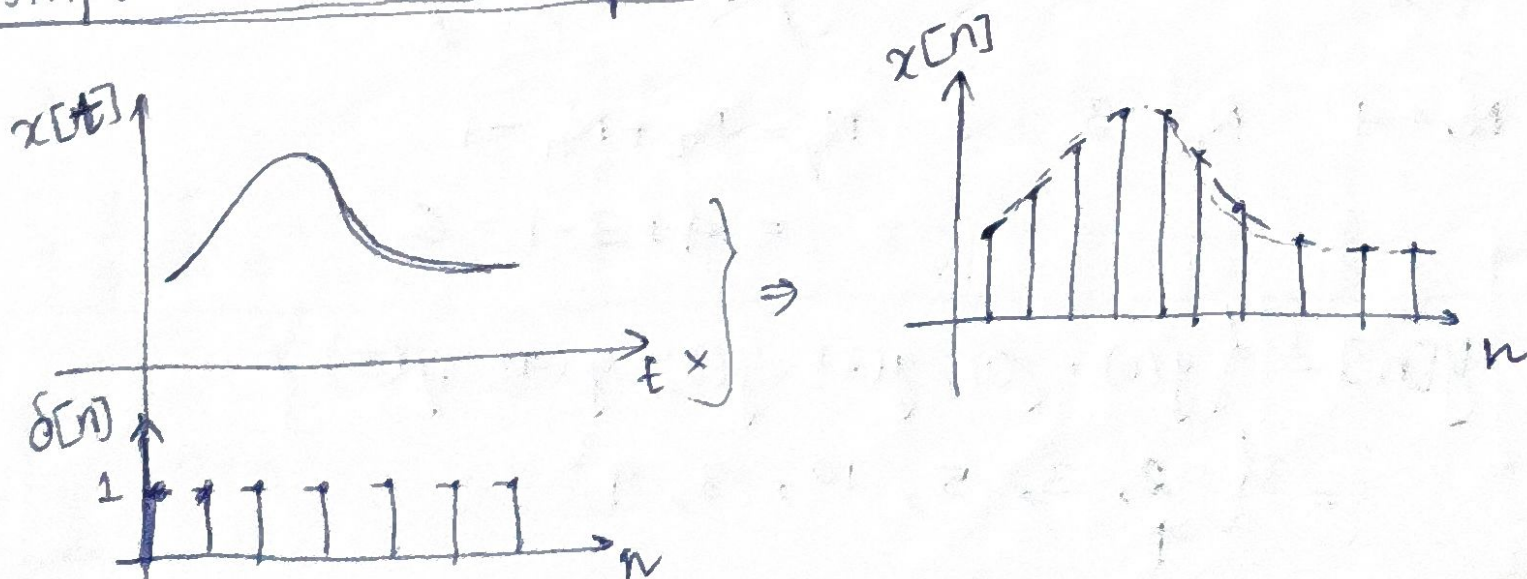
It states that when ~~we sample~~ a continuous analog signal is sampled to convert to discrete time signal, the sampling frequency (f_s) must be greater than ~~the~~ or equal to the twice of the maxm frequency (f_m) of the original signal, such that the original signal can be reconstructed from the sampled signal.

$$f_s \geq 2f_m$$

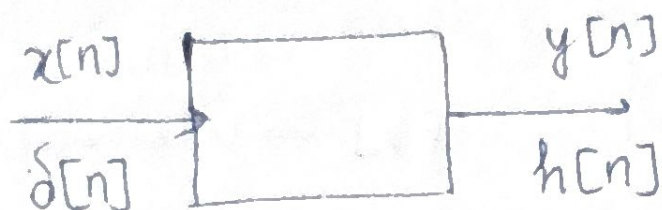
Nyquist rate:- It is the minimum sampling frequency to reconstruct the original signal from the sampled signal.

$$f_s = 2f_m$$

Impulse modulated signal



* Impulse Response



$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

$$\sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Q: $x[n] = u[n], h[n] = u[n]$

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} u[n] u[n-k]$$

$$= \sum_{k=0}^n u[n] u[n-k]$$

$$= \sum_{k=0}^n 1 \cdot 1 = (n+1) u[n]$$

Q:

$$x[n] = \{ \underset{\uparrow}{1}, 2, 3 \}$$

$$h[n] = \{ \underset{\uparrow}{2}, -1, 3 \}$$

$$LE_x = 0 \quad RE = 3$$

$$LE_y = LE_x + LE_h = 0$$

$$LE_h = 0 \quad RE = 2$$

$$RE_y = RE_x + RE_h = 3 + 2 = 5$$

$$N_x = 4 \quad N_h = 3$$

$$N_y = N_x + N_h - 1$$

$$= 4 + 3 - 1 = 6$$

$$y[n] = \{ y(0), y(1), y(2), y(3), y(4), y(5) \}$$

$$= \{ \underset{\uparrow}{2}, 3, 5, 10, 3, 9 \}$$

Tabular method

$x(n)$	$x(0)=1$	$x(1)=2$	$x(2)=2$	$x(3)=3$
$h(n)$				
$h(0)=2$	(2)	4	4	6
$h(1)=1$	-1	-2	-2	-3
$h(2)=3$	3	6	6	(9)

$$y[n] = \sum_{k=-\infty}^{\infty} x(k) * h(n-k)$$

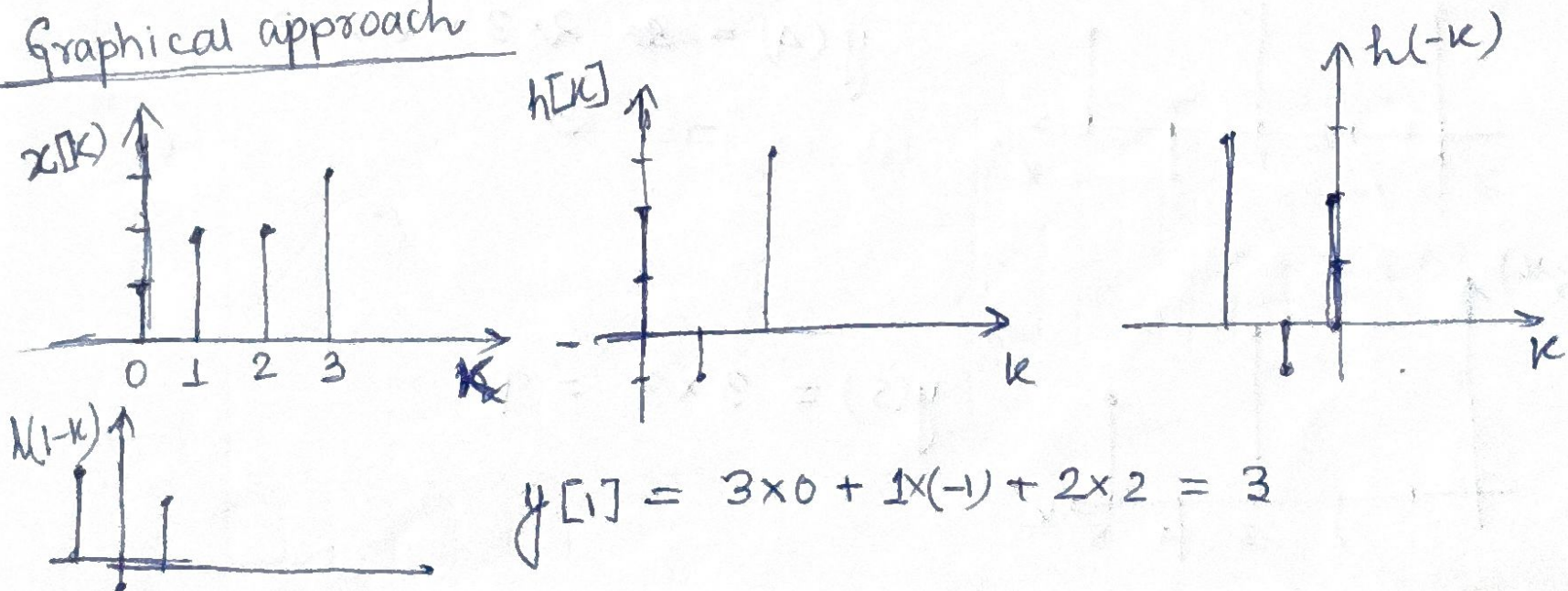
$$y(0) = \sum_{k=-\infty}^{\infty} x(k) h(-k) = x(0) h(0)$$

$$y(1) = \sum_{k=-\infty}^{\infty} x(k) h(1-k) = x(1) h(0) + x$$

Multiplication method:

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 2 & 2 & 3 \\
 & 2 & -1 & 3 \\
 \hline
 3 & 6 & 6 & 9 \\
 -1 & -2 & -2 & -3 & \times \\
 2 & 4 & 4 & 2 & \times \\
 \hline
 2 & 3 & 5 & 10 & 3 & 9
 \end{array} \\
 y[n] = \{ \uparrow \\
 2 & 3 & 5 & 10 & 3 & 9 \}
 \end{array}$$

Graphical approach



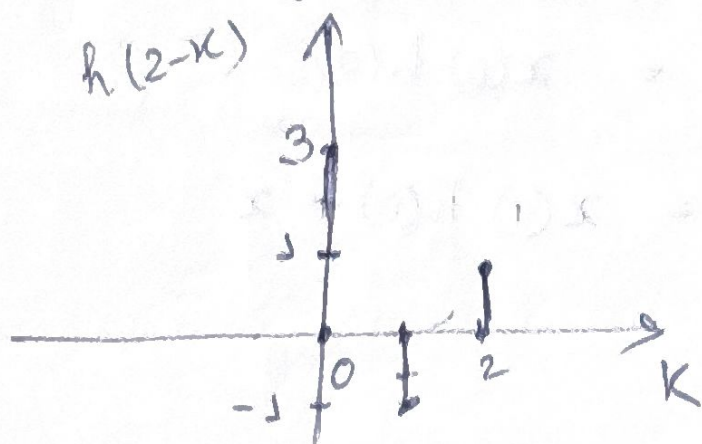
$$y[n] = x[n] * h[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

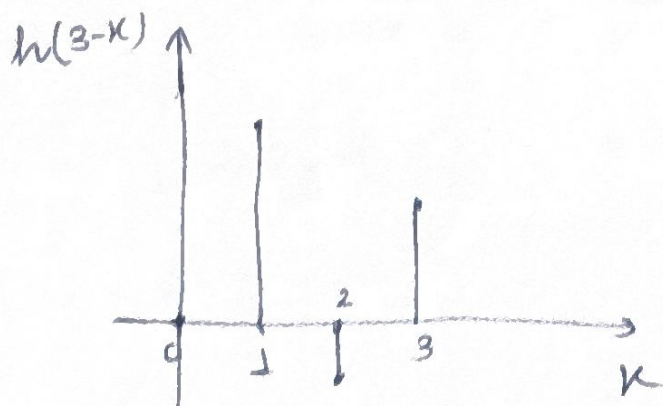
$$y[0] = \sum_{k=0} x[k] h[-k] = x[0] h[0]$$

$$y[1] = \sum_{k=0}^1 x[k] h[1-k] = x[0] h[1] + x[1] h[0]$$

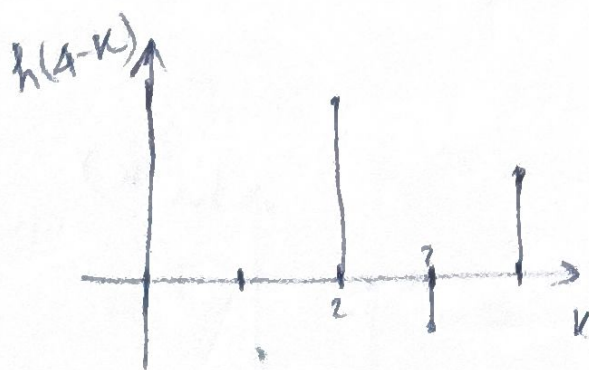
$$h(-k) \xrightarrow[\text{by 2}]{\text{RS}} h[-(k-2)] = h(2-k)$$



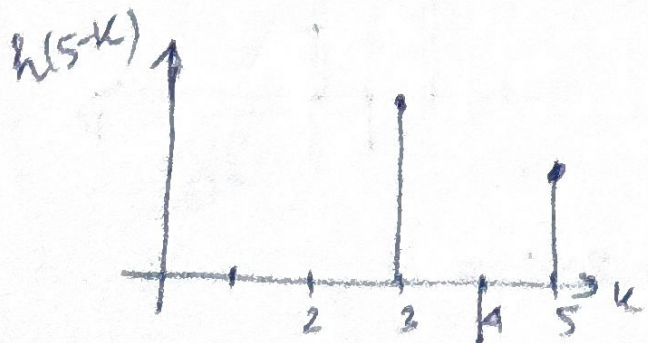
$$\begin{aligned} y[2] &= 3 \times 1 + (-1) \times 2 + 2 \times 2 \\ &= 3 - 2 + 4 \\ &= 5 \end{aligned}$$



$$\begin{aligned} y[3] &= 2 \times 3 + 2(-1) + 2 \times 3 \\ &= 10 \end{aligned}$$



$$\begin{aligned} y[4] &= 2 \times 3 + (-1) \times 3 \\ &= 3 \end{aligned}$$



$$y[5] = 3 \times 3 = 9$$

Prob 1:

$$h(n) = a^n u(n) \quad |a| < 1$$

$$x(n) = u(n)$$

$$y(n) = x(n) * h(n) = ?$$

$$= \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} u(k) a^{n-k} u(n-k)$$

$$= \sum_{k=0}^n a^{n-k}$$

$$= a^n + a^{n-1} + a^{n-2} + \dots + a^0$$

$$= 1 + \dots + a^{n-1} + a^{n-2} + a^n$$

$$= \frac{1(1 - a^{n+1})}{1-a} u[n]$$

Prob 2: $x(n) = \left\{ \underset{\uparrow}{1}, 2, -1 \right\}$ $h(n) = x(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} x(k) x(n-k)$$

$$LE_x = -1 \quad RE_x = 1$$

$$LE_h = -1 \quad RE_h = 1$$

$$LE_y = -1-1 = -2 \quad RE_y = 1+1 = 2$$

$$N_y = N_x + N_h - 1$$

$$= 3 + 3 - 1 = 5$$

$$y(n) = \left\{ \begin{array}{cccc} & 1 & 2 & -1 \\ & 1 & 2 & -1 \\ \hline & -1 & -2 & 1 \\ & 2 & 4 & -2 & x \\ \hline 1 & 2 & -1 & & x \\ \hline 1 & 4 & 2 & -4 & 1 \end{array} \right\}$$

Properties

1. $x(n) * h(n) = h(n) * x(n)$
2. $[x(n) * h_1(n)] * h_2(n) = x(n) * [h_1(n) * h_2(n)]$
3. $x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n)$

Q. $x(n) = h(n) = \delta(n)$

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \delta(k) \delta(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \delta(k) \sum_{k=-\infty}^{\infty} \delta(n-k)$$

$$= 1$$

$$\delta(k) = \begin{cases} 1, & k=0 \\ 0, & \text{else} \end{cases}$$

Q. $x(n) = \{ \underset{\uparrow}{0}, 0, 0, 1, 1, 1, 1 \}$

$$h(n) = \{ 1, 1, 0, 0, \underset{\uparrow}{0} \}$$

$$y(n) = x(n) * h(n)$$

Soln:

$$LE_y = 0 - 4 = -4$$

$$NE_y = 7 + 5 - 1$$

$$= 11$$

$$RE_y = 6 + 0 = 6$$

$x(n)$	$n=0$	$x(1)$	$x(2)$	$x(3)$	$x(4)$	$x(5)$	$x(6)$
$x(0)=0$	$=0$	$=0$	$=0$	$=1$	$=1$	$=1$	$=1$
$h(n)$							
$h(4)=1$	0	0	0	1	1	1	1
$h(3)=1$	0	0	0	1	1	1	1
$h(2)=0$	0	0	0	0	0	0	0
$h(1)=0$	0	0	0	0	0	0	0
$h(0)=0$	0	0	0	0	0	0	0

$$h(-k) \xrightarrow{L^*} h(k+4)$$

$$\xrightarrow{D^*} h(-4-k)$$

$$y[n] = \{0, 0, 0, 1, 2, 2, 2, 1, 0, 0, 0\}$$

$$y(-4) = \sum_{k=-\infty}^{\infty} x(k) h(-4-k)$$

$$= 0 \times 1 = 0$$

$$y(-3) = 0 \times 1 + 0 \times 1 = 0$$

$$y(-2) = 0 \times 1 + 0 \times 1 + 0 = 0$$

$$y(-1) = 0 + 0 + 0 + 1 \times 1 = 1$$

$$y(0) = 1 \times 1 + 1 \times 1 = 2$$

$$y(1) = 1 \times 1 + 1 \times 1 = 2$$

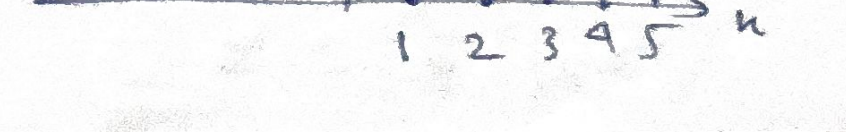
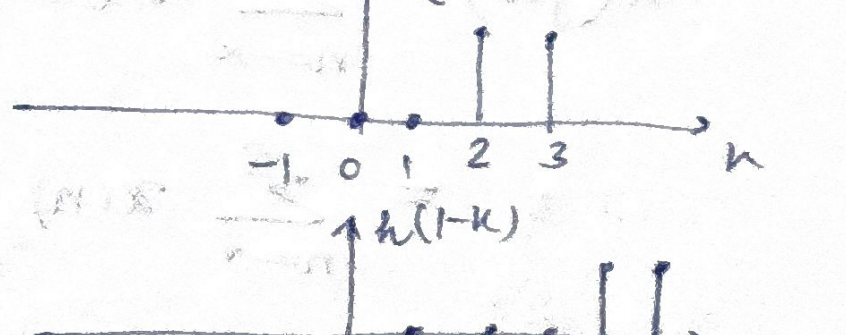
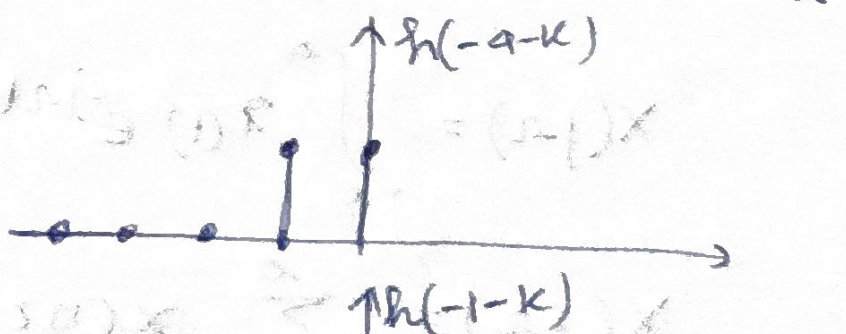
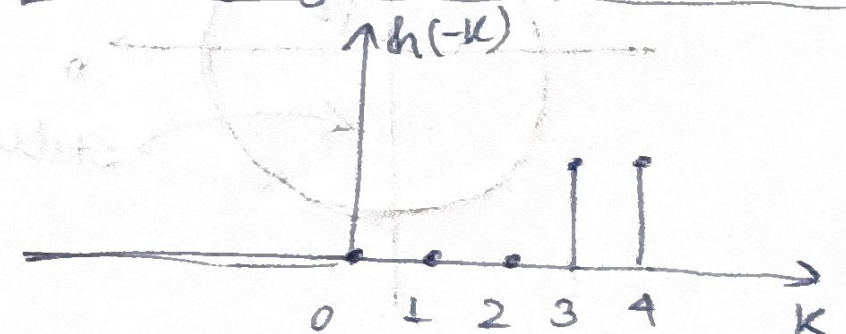
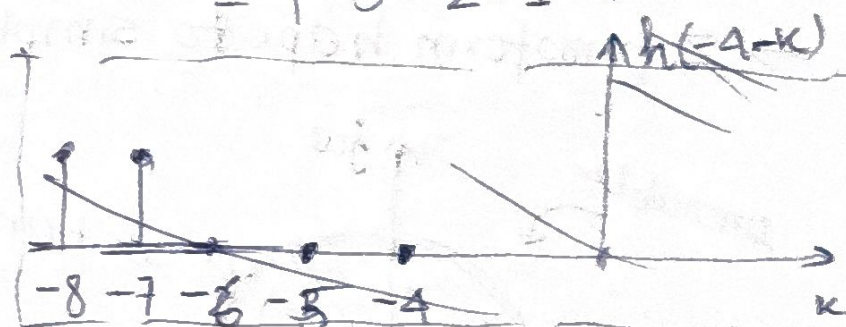
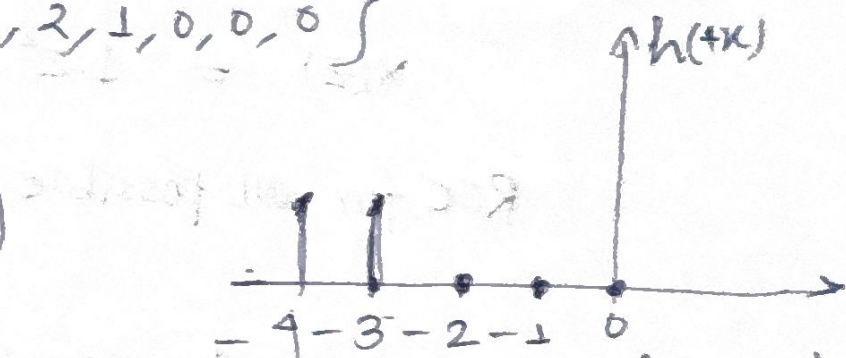
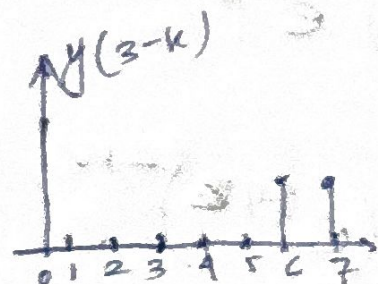
$$y(2) = 1 \times 1 + 1 \times 1 = 2$$

$$y(3) = 1 \times 1 = 1$$

$$y(4) = 0$$

$$y(5) = 0$$

$$y(6) = 0$$



* Z Transform

$$X(z) = Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$\text{eg: } x(n) = \{1, 2, 3, 4, 5\}$$

$$= \sum_{n=-2}^{\infty} x(n) z^{-n}$$

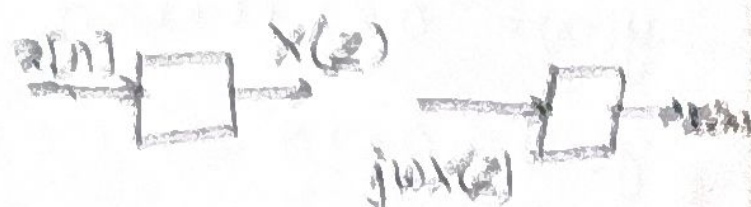
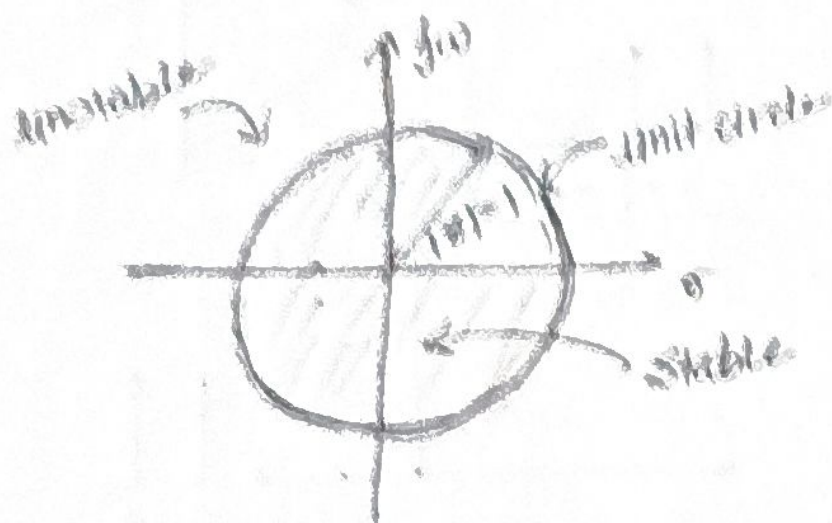
$$= x(-2) z^{-(-2)} + x(-1) z^{-(-1)} + x(0) z^{-0}$$

$$+ x(1) z^{-1} + x(2) z^{-2}$$

$$X(z) = 1 \cdot z^2 + 2z + 3 + 4z^{-1} + 5z^{-2}$$

Roc for all possible point in z plane exist (except $z=0$)

→ Z transform helps to simplify the difference eqn = stability



$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \rightarrow$$

$$X(j\omega) = \sum_{n=-\infty}^{\infty} x(nT) e^{-j\omega nT} \rightarrow \text{Discrete Time Sequences}$$

$$= \sum_{n=-\infty}^{\infty} x(n) (e^{j\omega T})^{-n}$$

1. $x[n] = \delta[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n] z^{-n}$$

$$= X(0) z^0$$

$$= \delta(0) = 1$$

ROC: Entire z-plane

2. $x[n] = \{0, 0, 1, 2, 3\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= 1 \times z^{-2} + 2 \times z^{-3} + 3 \times z^{-4}$$

ROC: Entire z-plane except $z=0$

3. $x[n] = \delta[n-k]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n-k] z^{-n}$$

$$= \delta[k-k] z^{-k}$$

$$= z^{-k}$$

ROC: Entire z-plane except $z=0$

4. $x[n] = \delta[n+k]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n+k] z^{-n}$$

$$= \delta[-k+k] z^{-k}$$

ROC: Entire z-plane except $z=\infty$

* Causal system $x[n] = u[n]$

Anticausal $\Rightarrow x_1[n] = u[-n-1]$

$$Z[x[n]] = X(z) = \sum_{n=-\infty}^{\infty} u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n$$

$$z^{-1} < 1$$

$$|z| > 1 : \text{ROC}$$

$$X(z) = \frac{1}{1 - z^{-1}}$$

$$Z[x_1[n]] = X_1(z) = \sum_{n=-\infty}^{\infty} u[-n-1] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} (z^{-1})^{n+1}$$

$$|z| < 1$$

$$= \frac{1}{1 - z^{-1}} \cdot z^{-1} = \frac{z^{-1}}{1 - z^{-1}}$$

1. $x[n] = \delta[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n] z^{-n}$$

$$= x[0] z^0$$

$$= \delta[0] = 1$$

ROC: Entire z-plane

2. $x[n] = \{0, 0, 1, 2, 3\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= 1 \times z^{-2} + 2 \times z^{-3} + 3 \times z^{-4}$$

ROC: Entire z-plane except $z=0$

3. $x[n] = \delta[n-k]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n-k] z^{-n}$$

$$= \delta[k-k] z^{+k}$$

$$= z^k$$

ROC: Entire z-plane
except $z=0$

4. $x[n] = \delta[n+k]$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n+k] z^{-n}$$

$$= \delta[-k+k] z^{-k}$$

ROC: Entire z-plane
except $z=\pm\infty$

* Causal system $\Rightarrow x[n] = u[n]$

Anticausal $\Rightarrow x[n] = u[-n-1]$

$$Z[x[n]] = X(z) = \sum_{n=-\infty}^{\infty} u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n$$

$$X(z) = \frac{1}{1 - z^{-1}}$$

$|z| > 1$: ROC

$$Z[x_1[n]] = X_1(z) = \sum_{n=-\infty}^{\infty} u[-n-1] z^{-n}$$

$$= \sum_{n=-\infty}^{-1} (z^{-1})^n$$

$$= \sum_{m=1}^{\infty} z^m = \frac{z}{1-z} = \frac{-1}{1-z^{-1}}$$

$|z| < 1$

2. $x(n) = a^n u(n) = a^n, n \geq 0$

ROC for Anticausal $\rightarrow$ interior to the circle

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n}$$

ROC for Causal $\rightarrow$ Exterior to the circle

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \frac{1}{1-a} \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{1}{1-az^{-1}}$$

ROC: $|az^{-1}| < 1$

ROC $\Rightarrow |z| > |a|$

$x(n) = -a^n u(-n-1)$

$$X(z) = \sum_{n=-\infty}^{\infty} -a^n u(-n-1) z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -a^n z^{-n} = \sum_{m=1}^{\infty} -a^{-m} z^m$$

$(za^{-1})^m < 1$

$$= \frac{-a^{-1}z}{1-a^{-1}z}$$

$|z| < |a|$

$$= \frac{1}{1-az^{-1}}$$

Q $x(n) = a^n u(n) + \beta^n u(-n-1)$

$$Z\{x(n)\} = X(z) = \frac{1}{1-az^{-1}} + \frac{1}{1-\beta z^{-1}}$$

ROC: $|z| > |a|$

ROC: $|z| < |\beta|$

CASE 1: $\alpha > \beta$

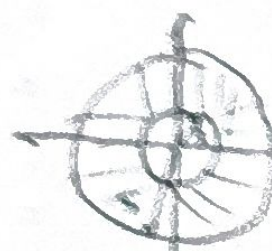
$5 > 3$
 $|z| > 5$
 $|z| < 3$



Z-transform will not exist

CASE 2: $\alpha < \beta$

$3 < 5$
 $|z| > 3$
 $|z| < 5$



ROC

$3 \leq |z| \leq 5$